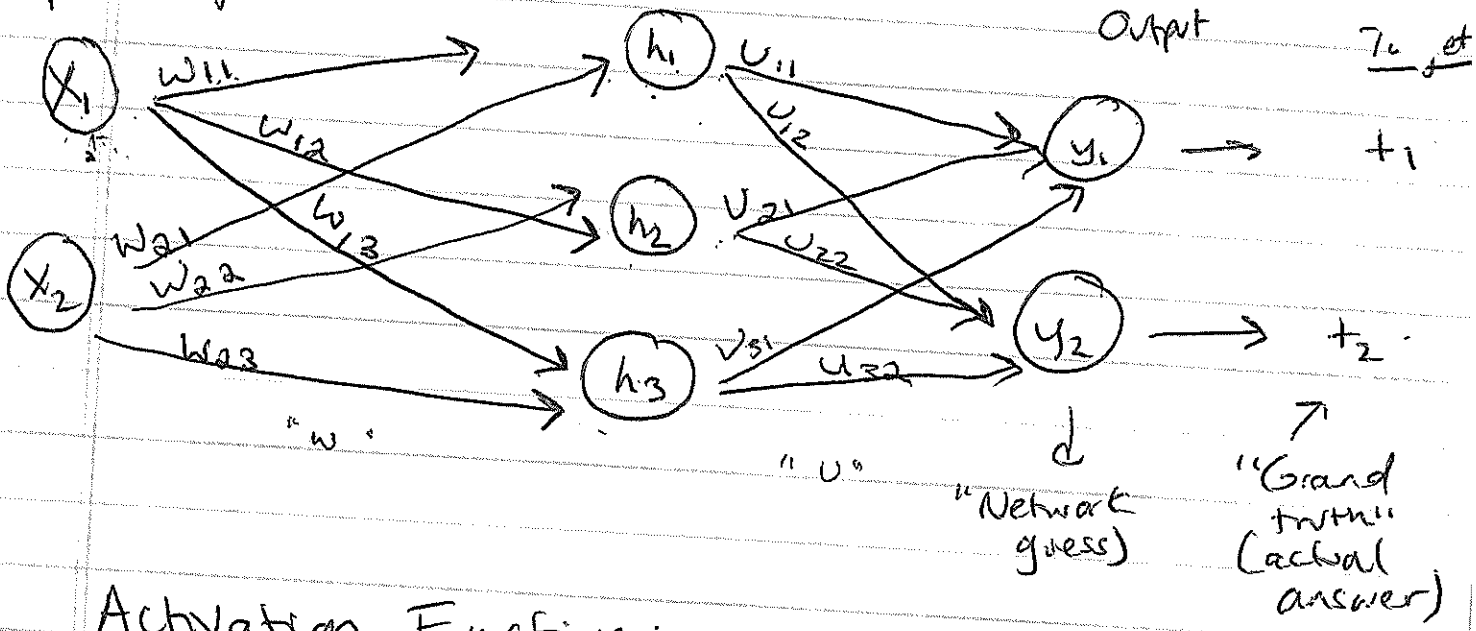
Basic Neural Network

$W_{prev_layer,\ next_layer}$ Hidden Layer

Input layer



Activation Function:

$$S(x) = \frac{1}{1+e^{-x}}$$

One-Hot Encoding

Dog 1 | Cat 2 | Lizard 3

Output layer

- y_1 → Probability of "dog" class
- y_2 → Probability of "cat" class
- y_3 → Probability of "lizard class"

Original grand truth → a number between 1-3

Change this to a probability vector of each class

Dog (1) → [1 0 0] ← len = # of classes

Cat (2) → [0 1 0]

Lizard (3) → [0 0 1]

Feed forward phase

① Multiply the input by the weight vector & add the bias term. This will give the "net" input of a single neuron/node "a"

a_1 = Net input of neuron 1. a_2 = net input of neuron 2 in a layer.

$a^{\text{layer}}_{\text{node}}$

$a_1^1 \rightarrow$ Neuron/node 1 in hidden layer 1
$a_2^1 \rightarrow$ Node 2, hidden layer 1
$a_1^2 \rightarrow$ Node 1, hidden layer 2
$a_2^2 \rightarrow$ Node 2, hidden layer 2

$$a_1^1 = x_1 \cdot w_{11} + x_2 \cdot w_{21} + b_1^1$$

↑ "Net" input for h_1.

② Activation Functions: Acts on the net input

Denote an activation function: $\sigma(\cdot)$

"Value" of $h_1 = \sigma(a_1^1)$

output/value

$$h_1 = \sigma(x_1 \cdot w_{11} + x_2 \cdot w_{21} + b_1^1) \quad (a_1^1)$$

$$h_2 = \sigma(x_1 \cdot w_{12} + x_2 \cdot w_{22} + b_2^1) \quad (a_2^1)$$

$$h_3 = \sigma(x_1 \cdot w_{13} + x_2 \cdot w_{23} + b_3^1) \quad (a_3^1)$$

↳ Input for our next layer

(3) Next: hidden to output layer

Net of $a_1^2 = h_1 \cdot U_{11} + h_2 \cdot U_{21} + h_3 \cdot U_{31} + b^2$

(output) $y_1 = \sigma(a_1^2) = \sigma(\quad)$

$y_2 = \sigma(a_2^2) = \sigma(h_1 \cdot U_{12} + h_2 \cdot U_{22} + h_3 \cdot U_{32} + b_2^2)$

END Feed forward

START: Error Calculation (Whole Network)

y_1 & y_2 are our "output"

t_1 & t_2 are our "targets" / ground truth

Actual values we want our network to produce for our input sample X_1, X_2

After feed forward, we calculate the error of our network:

(MSE) loss/error/cost; output of network node i; target i

L2 norm loss : $L = \frac{1}{2} \sum (y_i - t_i)^2$

$y_1 \rightarrow 0.2 \quad t_1 = 0$

$y_2 \rightarrow 0.8 \quad t_2 = 1$

$$\frac{(0.2-0)^2 + (0.8-1)^2}{2}$$

End Error Calc

Start Backpropagation start with output → hidden layer

Single weight U_{ij}

j = index of output node

i = index of input node

derivative of loss w.r.t. parameter

$$\frac{\partial L}{\partial U_{ij}} = \frac{\partial L}{\partial y_j} * \frac{\partial y_j}{\partial a_j^2} \cdot \frac{\partial a_j^2}{\partial U_{ij}}$$

i = prev node
j = next node

Backprop for the output layer → hidden layer

$$\frac{d}{du_{ij}} = \frac{dL}{dy_j} * \frac{dy_j}{da_j^2} * \frac{da_j^2}{du_{ij}}$$

A. $\frac{dL}{dy_j}$ = (partial) Derivative of the loss function w.r.t. to output node j. = Partial derivative of our loss function.

Loss function: $L = \frac{1}{2}\sum(y_i - t_i)^2$

$= (y_j - t_j)$ ✓

B. $\frac{dy_j}{da_j^2}$ = Derivative of our activation function.

$S(x) = \frac{1}{1+e^{-x}}$ → This function turns a_j^2 into y_j

$\frac{dy_j}{da_j^2}\ \sigma(a_j^2)(1-\sigma a_j^2) = y_j(1-y_j)$ ✓

C. $\frac{da_j^2}{du_{ij}} = a^2 = h * u + b$ (given hidden layer value; given weight; bias)

$= h_i u_{ij} + b$

$f'(h_i u_{ij} + b)$

h_i ← Input multiplied by that particular weight.

✓

$$\frac{dL}{du_{ij}} = \frac{dL}{dy_j} * \frac{dy_j}{da_j^2} * \frac{da_j^2}{du_{ij}} = (y_j - t_j)(y_j)(1-y_j)h_i$$

(5)

Next: Hidden Layer Backprop

$$\frac{dL}{dw_{ij}} = \frac{dL}{dh_j} * \frac{dh_j}{da_j^1} * \frac{da_j^1}{dw_{ij}}$$ ← weights connecting hidden layer & inputs.

B. $$\frac{dh_j}{da_j^{(1)}} = \sigma(a_j^1)(1-\sigma a_j^1) = h_j(1-h_j)$$ ✓

C. $$\frac{da_j^1}{dw_{ij}} = f'(\underbrace{x}_{\text{inputs}} \cdot \underbrace{w}_{\text{weights}} + \underbrace{b}_{\text{bias}}) = x_i$$ ✓

A. $$\frac{dL}{dh_j} \rightarrow ? \rightarrow$$ We have no hidden target

h_1 example:

$$\frac{dL}{dh_1} = \frac{dL}{dy_1}\frac{dy_1}{da_1^{(2)}}\frac{da_1^{(2)}}{dh_1} + \frac{dL}{dy_2}\frac{dy_2}{da_2^2}\frac{da_2^2}{dh_1}$$

$(y_1 - t_1)$ $y_1(1-y_1)$ u_{11} $(y_2 - t_2)$ $y_2(1-y_2)$ u_{12}

So $$\frac{dL}{dw_{11}} = \underbrace{\left[(y_1-t_1)y_1(1-y_1)u_{11} + (y_2-t_2)(y_2)(1-y_2)u_{12}\right]}_{\frac{dL}{dh_1}} \underbrace{h_1(1-h_1)}_{\frac{da_1^1}{dh_1}} \underbrace{x_1}_{\frac{dh_1}{da_1^1}\frac{da_1^1}{dw_{11}}}$$

General form of backprop equation:

$$\frac{dL}{dw_{ij}} = \sum_K (y_K - t_K) y_K (1 - y_K) v_{jK} h_j (1 - h_j) x_i$$

K output nodes

i = layer before j = layer after

$$\frac{dL}{dw_{ij}} = \delta_j * x_i$$

For a hidden layer:

$$\delta_j = \sum_K \delta_K w_{jK} y_j (1 - y_j) x_i$$

- $\sum_K$: Layer we already backprop'd before
- δ_K: $(y - t) y (1 - y)$ weights
- w_{jK}: weights connecting hidden to output
- y_j: output
- x_i: input